

Orientation Examples

1 Orientation Formula Review

Orientation Test

For three points

$$P = (p_x, p_y), \quad Q = (q_x, q_y), \quad R = (r_x, r_y),$$

$$\text{orient}(P, Q, R) = (q_x - p_x)(r_y - p_y) - (q_y - p_y)(r_x - p_x).$$

The sign tells the turn direction:

Sign	Meaning
$\text{orient} > 0$	Left turn, counterclockwise.
$\text{orient} < 0$	Right turn, clockwise.
$\text{orient} = 0$	Collinear points.

2 Left-Turn Example

Problem

Let

$$P = (1, 1), \quad Q = (4, 2), \quad R = (5, 5).$$

Determine the turn made by

$$P \rightarrow Q \rightarrow R.$$

Solution

First compute the two vectors from P :

$$\overrightarrow{PQ} = (4 - 1, 2 - 1) = (3, 1),$$

$$\overrightarrow{PR} = (5 - 1, 5 - 1) = (4, 4).$$

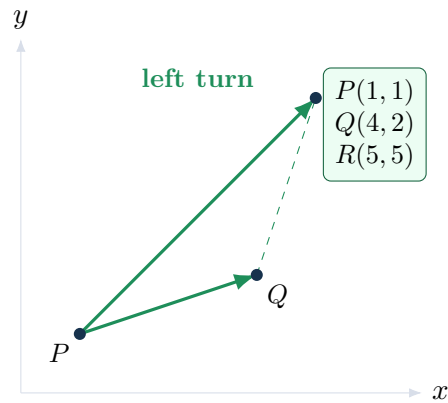
Now compute the cross product:

$$\text{orient}(P, Q, R) = 3(4) - 1(4) = 12 - 4 = 8.$$

Since

$$8 > 0,$$

the points form a left turn.



Answer

$\text{orient}(P, Q, R) = 8 > 0$, so the turn is left.

3 Right-Turn Example

Problem

Let

$$P = (1, 1), \quad Q = (4, 4), \quad R = (5, 2).$$

Determine the turn made by

$$P \rightarrow Q \rightarrow R.$$

Solution

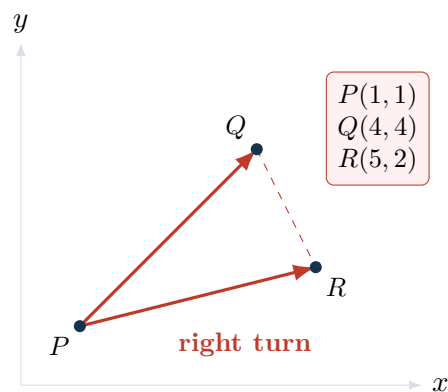
$$\begin{aligned} \overrightarrow{PQ} &= (4 - 1, 4 - 1) = (3, 3), \\ \overrightarrow{PR} &= (5 - 1, 2 - 1) = (4, 1). \end{aligned}$$

$$\text{orient}(P, Q, R) = 3(1) - 3(4) = 3 - 12 = -9.$$

Since

$$-9 < 0,$$

the points form a right turn.



Answer

$$\text{orient}(P, Q, R) = -9 < 0, \quad \text{so the turn is right.}$$

4 Collinear Points

Collinearity Test

If

$$\text{orient}(P, Q, R) = 0,$$

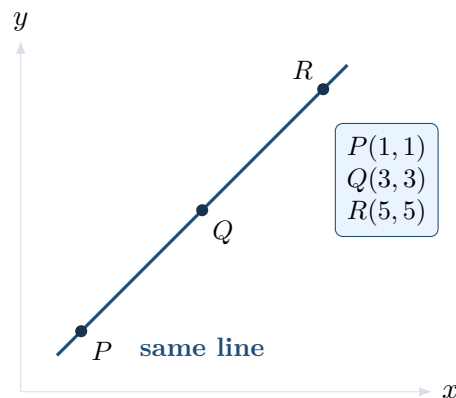
then P, Q, R lie on the same straight line.

Example:

$$P = (1, 1), \quad Q = (3, 3), \quad R = (5, 5).$$

$$\overrightarrow{PQ} = (2, 2), \quad \overrightarrow{PR} = (4, 4).$$

$$\text{orient}(P, Q, R) = 2(4) - 2(4) = 8 - 8 = 0.$$



5 Collinear Rule for Convex Hulls

When several points have the same direction from a pivot point, their orientation with the pivot is zero. In convex hull algorithms, the usual rule is:

Keep the Farther Point

If two candidate points have the same polar angle from the pivot, keep the farther point and discard the nearer point.

Let the pivot be

$$P_0 = (0, 0),$$

and let

$$A = (2, 2), \quad B = (4, 4).$$

Both points lie on the same ray from P_0 , so

$$\text{orient}(P_0, A, B) = 0.$$

Compare squared distances from P_0 :

$$d^2(P_0, A) = 2^2 + 2^2 = 8,$$

$$d^2(P_0, B) = 4^2 + 4^2 = 32.$$

Since

$$32 > 8,$$

keep B and discard A .

Why Discard the Nearer Point?

The nearer point lies inside the same boundary direction. Keeping it may place an interior collinear point on the convex hull, which is usually not desired.

6 Final Review

Orientation Examples Summary

$\text{orient} > 0 \Rightarrow$ left turn.
 $\text{orient} < 0 \Rightarrow$ right turn.
 $\text{orient} = 0 \Rightarrow$ collinear.
For same-angle hull candidates, keep the farther point.